WorksheetDownload.com

10 CONCEPTUAL Q&A DATA ENGINEERING · PRACTICE WORKSHEET

CONCEPTUAL SERIES
NumPy

Name: ______________ Cohort/Batch: ______________ Date: ______________ Score: ______________

NUMPY PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Data Engineering

TOTAL: 10 QUESTIONS

Q1 What is NumPy and what problem in Data Engineering does it solve? *Threat Model*

Q2 What are the core components or concepts in NumPy? *Architecture*

Q3 How does NumPy compare to its main alternatives? *Configuration*

Q4 How do you get started with NumPy for a new project? *Hardening*

Q5 What configuration options most affect NumPy's behavior in production? *SSO / IdP*

Q6 How does NumPy handle reliability and high availability? *Observability*

Q7 What observability does NumPy provide out of the box? *Lifecycle*

Q8 What are common performance bottlenecks in NumPy? *Compliance*

Q9 What security best practices apply when running NumPy? *Testing*

Q10 What mistakes do teams commonly make when adopting NumPy? *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 NumPy is a tool or platform

Mitigation

NumPy is a tool or platform that automates or simplifies a specific concern in the Data Engineering domain, reducing manual effort and operational complexity for engineering teams.

A2 NumPy has key building blocks that

Primitives

NumPy has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A3 NumPy trades off simplicity against feature

Hardening

NumPy trades off simplicity against feature richness differently than its alternatives. Choose NumPy when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A4 Install NumPy via its package manager

Gotchas

Install NumPy via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A5 Production-critical settings typically include

Federation

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A6 NumPy provides HA through leader election,

Observability

NumPy provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A7 NumPy exposes metrics (Prometheus endpoint

Automation

NumPy exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A8 Performance bottlenecks typically stem from

Governance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A9 Run NumPy with the principle of

Assessment

Run NumPy with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A10 Common pitfalls include skipping the

Optimization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.